

SECURITY TEST CASES

SEC-001: SQL Injection Prevention

Priority: P1 - High

Test Input	Field	Expected Result	Actual Result	Status
' OR '1'='1	Login Email	Error or no effect, not logged in	Not logged in	☒ Pass ☐ Fail
admin'--	Login Email	Error or no effect	No effect	☒ Pass ☐ Fail
1'; DROP TABLE equipment--	Search Box	Error or no results, table not dropped	No results	☒ Pass ☐ Fail
<script>alert('XSS')</script>	Equipment Name	Escaped, not executed	Not executed	☒ Pass ☐ Fail

SEC-002: XSS (Cross-Site Scripting) Prevention

Priority: P1 - High

Test Input	Field	Expected Result	Actual Result	Status
<script>alert('XSS')</script>	Request Purpose	Text displayed as-is, not executed	Text displayed as-is, not executed	☒ Pass ☐ Fail
	Equipment Description	Image tag not rendered	Image tag not rendered	☒ Pass ☐ Fail
javascript:alert('XSS')	Any URL field	Not executed	Note executed	☒ Pass ☐ Fail

SEC-003: Authentication & Authorization

Priority: P0 - Critical

Test Input	Field	Expected Result	Actual Result	Status
1	Logout, try to access /admin/equipment directly	Redirected to login	The system redirected to login	☒ Pass ☐ Fail
2	Login as student, manually navigate to /admin URL	Access denied / redirected	Access denied	☒ Pass ☐ Fail
3	Try to approve request via API as student	403 Forbidden	403 Forbidden	☒ Pass ☐ Fail
4	Try to delete equipment via API as faculty	403 Forbidden	403 Forbidden	☒ Pass ☐ Fail
5	Session expires, try to perform action	Redirected to login	The system redirected to login	☒ Pass ☐ Fail

SEC-004: Password Security

Priority: P1 - High

Test Case	Expected Result	Status
Check if passwords are hashed in database	Yes, not plain text	☒ Pass ☐ Fail
Try password less than 8 characters	Error: "Minimum length is 8 characters"	☒ Pass ☐ Fail
Try password: "password"	Error: "Password must have uppercase, lowercase, number, and special character"	☒ Pass ☐ Fail
Try password with 8+ chars, numbers, special chars	Accepted	☒ Pass ☐ Fail
Verify password not shown in any logs	Not visible	☒ Pass ☐ Fail